Practice 1 - Setting up the Stretch Robot Simulator

Robotics and Cyber-Physical Systems
School of Sciences and Engineering
Tecnológico de Monterrey

1 Introduction

The Hello Robot Stretch is a mobile manipulator for research and assistive robotics. This practice introduces the Stretch MuJoCo Digital Twin—a high-fidelity physics simulation for developing robotics algorithms.



Figure 1: Stretch robot operating in the MuJoCo simulation environment

Objectives:

- Install and verify the base simulator
- Install and verify RoboCasa kitchen environments
- Toggle between environments via configuration
- Complete a block stacking challenge using teleoperation

2 Requirements

- Windows, macOS, or Linux laptop
- Git
- `uv` package manager
- Code editor (VS Code recommended)
- **Optional:** Xbox-compatible gamepad
- 5+ GB free disk space

3 Setup

Video Tutorial: `https://youtu.be/FLPB3TQcUoI`

3.1 Install uv Package Manager

Windows:

```
powershell -c "irm https://astral.sh/uv/install.ps1 | iex"
```

macOS/Linux:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Restart your terminal after installation.

3.2 Clone Repository

```
git clone https://github.com/SilentSammy/stretch_mujoco_digital_twin --recurse-submodules
cd stretch_mujoco_digital_twin
```

Linux users: If you get build errors, install:

```
sudo apt install build-essential python3-dev linux-headers-$(uname -r)
```

4 Part 1: Verify Base Simulation

Run the simulator:

```
uv run teleop_demo.py
```

You should see:

- Terminal: `=== Running on simulation backend ===`
- MuJoCo viewer with Stretch robot
- Two colored blocks (blue box and red cylinder)

Important: Before controlling the robot, click away from the MuJoCo viewer (e.g., click the taskbar or another window). This prevents keyboard inputs from triggering simulator functions. The controls will still work when the window is unfocused.

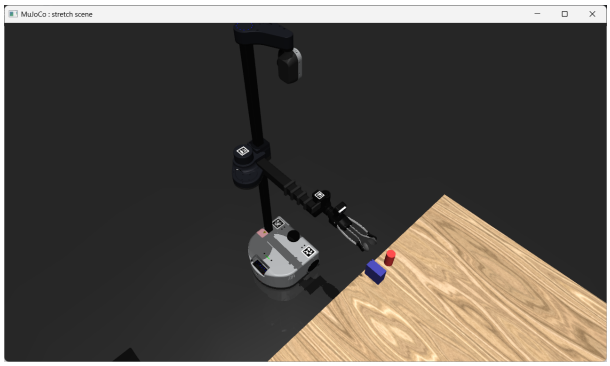


Figure 2: Default simulation environment

Press **1-5** to cycle through camera views (see Part 4 for details). Press **Ctrl+C** to exit.

5 Part 2: Install RoboCasa Environment

Install RoboCasa dependencies:

```
uv pip install -e ".[robocasa]"
uv pip install -e "robocasa@third_party/robocasa"
uv pip install -e "robosuite@third_party/robosuite"
uv run third_party/robosuite/robosuite/scripts/setup_macros.py
uv run third_party/robocasa/robocasa/scripts/setup_macros.py
uv run third_party/robocasa/robocasa/scripts/download_kitchen_assets.py
```

Note: Asset download is 5 GB and takes some time.
Verify `stretch_toolkit/robocasa_config.json` has `"enabled": true`, then run:

```
uv run teleop_demo.py
```

You should see a kitchen environment with cabinets, counters, and appliances.



Figure 3: RoboCasa kitchen scene

6 Part 3: Disable RoboCasa

Edit `stretch_toolkit/robocasa_config.json` and set `"enabled": false`:

```
{
  "enabled": false,
  "task": "PnPCounterToCab",
  "layout": 0,
  "style": 0
}
```

Run `uv run teleop_demo.py` again. You should see the simple block environment.
Leave RoboCasa disabled for Part 4.

7 Part 4: Block Stacking Challenge

Task: Stack one block on top of the other using teleoperation.

7.1 Stretch's 9 Degrees of Freedom

1. Base Translation
2. Base Rotation
3. Lift

4. Arm
5. Wrist Yaw
6. Wrist Pitch
7. Wrist Roll
8. Gripper
9. Head Pan/Tilt

7.2 Keyboard Controls

Keys	Action
W / S	Base forward/backward
D / A	Base rotate right/left
Z / X	Lift up/down
V / C	Arm extend/retract
M / N	Gripper open/close
L / J	Wrist yaw
U / O	Wrist roll
K / I	Wrist pitch
H	Toggle head/wrist mode

Table 1: Primary keyboard mappings (see `teleop_mappings.json` for gamepad)

7.3 Camera Views

Press keys **1–5** to toggle individual camera feeds on or off:

1. Head RGB
2. Head Depth
3. Wrist RGB
4. Wrist Depth
5. Head Navigation Cam

Note: Each active camera reduces simulation performance. Disable unused cameras if the simulation feels slow.

Task: During the block stacking challenge, experiment with at least two different camera views to assist your manipulation.

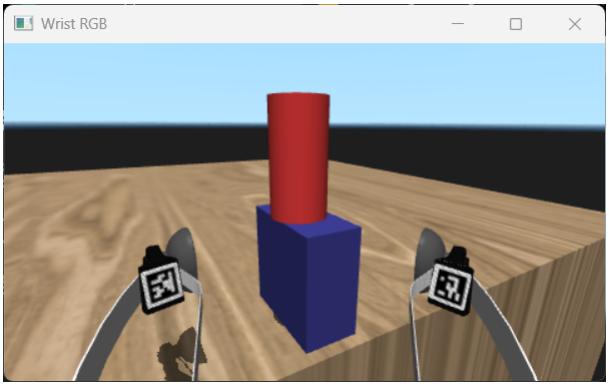


Figure 4: Target outcome: one block stacked on top of the other

Tips: Use different camera angles for better depth perception. Take your time.

8 Evidence and Submission

Submit a PDF report (1-2 pages) containing:

- **Link to a short video:** Move each joint and briefly explain what it does in your own words as it moves.
- Screenshots of the RoboCasa kitchen environment (Part 2) and the stacked blocks result (Part 4).
- A short written description of each of the robot's 9 degrees of freedom and its corresponding control key.

File naming: `LastName_FirstName_Practice1.pdf`

9 Troubleshooting

Simulator won't start: Run `uv sync` then retry
Controls not responding: Click the MuJoCo viewer window to focus it
Physics issues: Restart simulation with Ctrl+C

9.1 Resources

- `https://github.com/SilentSammy/stretch_mujoco_digital_twin`
- `https://mujoco.readthedocs.io/`
- `https://hello-robot.com/stretch-3`

10 Conclusion

You've installed the Stretch simulator, explored both simple and kitchen environments, and completed a manipulation task. Future practices will cover programmatic control, computer vision, and autonomous planning.

Good luck with the block stacking challenge!